

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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 ID :
 Section : 03

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

Then draw the tree

```
yiya@secr2043:~$ nano yiya.sh
yiya@secr2043:~$ chmod +x yiya.sh
yiya@secr2043:~$ ./yiya.sh
```

```

yiya@secr2043:~$ tree ~
/home/yiya
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
`-- yiya.sh

```

- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display `c` program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2 interactive.sh
#!/bin/bash
read -p "Enter file extension(s): " ext
echo "Files with .$ext extension:"
find . -type f -name ".*$ext"
count=$(find . -type f -name ".*$ext" | wc -l)
echo "Number of .$ext files: $count"

yiya@secr2043:~$ nano interactive.sh
yiya@secr2043:~$ chmod +x interactive.sh
yiya@secr2043:~$ ./interactive.sh
Enter file extension(s): c
Files with .c extension:
./drinks/accord.c
./drinks/proton.c
./drinks/civic.c
./cars/accord.c
./cars/proton.c
./cars/civic.c
Number of .c files: 6

```

2. The following Figure 1 illustrates a tree structure of some directories and files.

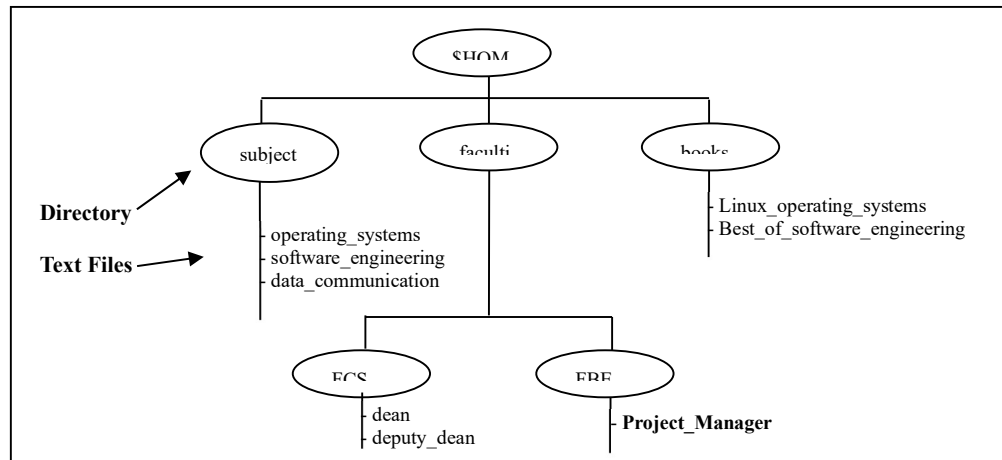


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```

GNU nano 7.2                                     yiya2a.sh
#!/bin/bash

# Create parent directories
mkdir -p subject
mkdir -p books
mkdir -p faculty/FCS
mkdir -p faculty/FBE

# Create text files in subject/
echo "operating systems" > subject/operating_systems
echo "software engineering" > subject/software_engineering
echo "data communication" > subject/data_communication

# Create text files in books/
echo "Linux operating systems" > books/Linux_operating_systems
echo "Best of software engineering" > books/Best_of_software_engineering

# Create text files in faculty/FCS/
echo "dean" > faculty/FCS/dean
echo "deputy dean" > faculty/FCS/deputy_dean

# Create text file in faculty/FBE/
echo "Project Manager" > faculty/FBE/Project_Manager
  
```

```

yiya@secr2043:~$ nano yiya2a.sh
yiya@secr2043:~$ chmod +x yiya2a.sh
yiya@secr2043:~$ ./yiya2a.sh

```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--



- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

GNU nano 7.2 yiya2c.sh
#!/bin/bash

# Change permissions
chmod 755 subject
chmod 440 books/Best_of_software_engineering
chmod 605 faculty/FCS
chmod 026 faculty/FBE/Project_Manager

```

```

yiya@secr2043:~$ nano yiya2c.sh
yiya@secr2043:~$ chmod +x yiya2c.sh
yiya@secr2043:~$ ./yiya2c.sh
yiya@secr2043:~$ ls -ld subject books/Best_of_software_engineering faculty/FCS faculty/FBE/Project_Manager
-r--r----- 1 yiya yiya 29 Jun 18 16:47 books/Best_of_software_engineering
-----w-rw- 1 yiya yiya 16 Jun 18 16:47 faculty/FBE/Project_Manager
drw----r-x 2 yiya yiya 4096 Jun 18 16:47 faculty/FCS
drwxr-xr-x 2 yiya yiya 4096 Jun 18 16:47 subject

```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr-xr-x
Best_of_software_engineering	-r--r-----
FCS	drw----r-x
project_manager	-----w-rw-

End of Lab 3

*** *All the Best for Final Exam* ***